

Generative POPs: Automating Character-Based Promotions on Digital Signage in Retail Stores

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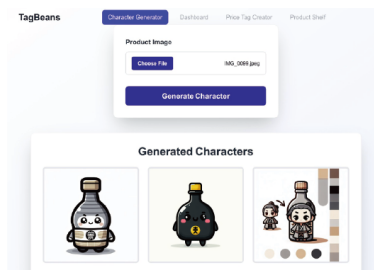
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(a) Photograph the product to be recommended using the application



(b) Select one of the automatically generated characters based on the product



(c) Display it on the digital signage, with labels shown in Japanese in the images.

Figure 1: An overview of the workflow from product image capture to character generation, selection, and POP display on digital signage.

Abstract

In this study, we propose a product recommendation system using AI-generated characters. Store staff can capture or select a product image, and multiple characters reflecting its features are automatically generated. A selected character is then displayed on shelf-edge digital signage. The system reduces the workload of creating promotional materials and enables flexible promotional expressions. An experiment with university campus convenience store staff suggested that generative AI-based flexible expressions can be effectively applied to real store operations.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

Generative AI, Digital Signage, Product Recommendation, Retail Technology, Human-Computer Interaction

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1 Introduction

In retail stores, promotional visuals such as posters and POP (Point-of-Purchase) displays are widely used to attract customers. These materials are usually created manually by store staff, resulting in heavy workloads and dependence on individual design skills. Recent advances in generative AI (Artificial Intelligence) enable high-quality content creation, including automatic poster generation [Lin et al. 2023] and layout generation based on product content and visual features [Guerreiro et al. 2024; Horita et al. 2024]. While these studies demonstrate the potential of AI-driven content generation, adoption in retail remains limited due to challenges in integration and usability.

We propose an integrated system that supports the entire process from AI-based POP generation to real-time display on digital signage. The system automatically generates character-based POPs tailored to each product image and seamlessly displays them in stores. We evaluated the system in a workflow simulating real store operations through a user study with six staff members. This paper describes the system's design and implementation, and reports on its practicality and potential for flexible, appealing promotion.

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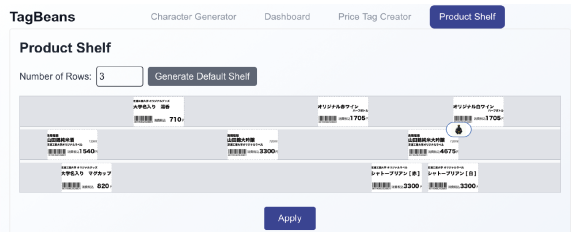


Figure 2: Web application interface reflecting the store shelf layout, where users can select a price tag to assign a character

2 System Overview

The proposed system enables store staff to create and operate promotional POP displays without specialized design expertise (Fig. 1). The core idea is to automatically generate character-based POPs tailored to individual products and seamlessly integrate them into digital signage. Store staff photograph the target product using the web application (Fig. 1a). Based on the captured image, three candidate characters are generated using OpenAI’s DALL-E 3 (Fig. 1b). Once a preferred character is selected, an animation is created using Runway’s Gen-4 Turbo. Each POP includes a “Recommended” label to highlight the product. Within the web interface reflecting the store’s shelf layout (Fig. 2), users assign the character to a price tag position, and the POP is then displayed on in-store digital signage (Fig. 1c). In this study, three shelf-edge digital signage units were installed in the price tag area to display the POPs (Fig. 3).

3 User Study

To evaluate the system’s effectiveness, we conducted an in-store experiment with six staff members from a university convenience store. A 90 cm-wide shelf-edge digital signage was installed in the price tag area of the store shelves, and a tablet was used to operate the web application. Participants freely selected a product photographed it, chose a character generated from its features, and displayed it at a designated price tag position. Following the experience, each participant completed a questionnaire and a semi-structured interview.



Figure 3: Three shelf-edge digital signage units installed on three tiers of shelves to display POPs.

Table 1: Average questionnaire scores (five-point Likert scale)

Questionnaire Item	Average Score
Experience with generative AI	1.67
Experience with creating promotional materials	3.00
Perceived usefulness of the system	5.00

3.1 Evaluation Items and Method

The questionnaire asked participants to rate three items on a five-point Likert scale: Q1: Have you ever used “generative AI” such as ChatGPT or image generation AI?; Q2: Have you ever created promotional materials such as POP displays or posters?; and Q3: Do you find this system useful?

In addition, semi-structured interviews gathered participants’ opinions on ease of operation, flexibility of expression, compatibility with existing tasks, and suggestions for improvement.

4 Results

4.1 Questionnaire Results

Table 1 shows the average scores for each item. Four out of six participants indicated no prior experience with generative AI. Experience in creating promotional materials varied with participants’ work backgrounds. All participants strongly agreed (5 on the Likert scale).

4.2 Interview Results

Interviews revealed several positive impressions, such as “It draws attention to the product,” “The movement catches the eye,” and “AI-generated POPs save time compared to handwritten ones.” At the same time, participants also suggested improvements, including “It would be better if the position of price tags and the colors or sizes of prices could be changed more freely” and “The selection of only three characters feels limited.”

5 Conclusion and future work

This paper presented a system that uses generative AI to automatically create characters based on product features and display them on shelf-edge digital signage. An in-store experiment with university campus convenience store staff demonstrated that the system is both useful and intuitive, even for participants with little prior experience using generative AI.

Interview revealed positive evaluations such as “It draws attention” and “It makes the product easier to understand,” along with suggestions including “It would be better if the color and size of prices could also be changed” and “It would be helpful if the display position of price tags could be adjusted.” These findings indicate that the system can address practical challenges in retail environments, though further flexibility and customization are needed.

This system has the potential to reduce the time and effort required to individually create and install promotional materials, while enabling the easy introduction of visually appealing promotional expressions. Although participants rated the system as easy to use, this may have been influenced by the short trial period. In future work, we will focus on improving usability and investigating the impact of long-term use on sales performance.

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